MATH2921 YEAR 2023 Quiz 1 ①

Solution

Q1. Solution: $x^2+y^2-z=1$

$$z = x^2+y^2-1.$$

$$f(x,y) = \boxed{x^2+y^2-1.}$$

$$x^2+y^2-z-1=0.$$

$$\nabla(x^2+y^2-z-1) = (2x, 2y, -1)$$

At $(1,0,0)$, $x=1$, $y=z=0$.

Normal: $(2,0,-1)$.

It should be

$$(-1)\cdot(2,0,-1) = (-2,0,1)$$

$a=-2$, $b=0$. $a+b=\boxed{-2.}$

②

Q2. Solution:

$$\nabla f = (2x-2, -2y+2)$$

$\nabla f = \vec{0}$: $x=y=1$.

$f_x = 2x-2$, $f_y = -2y+2$.

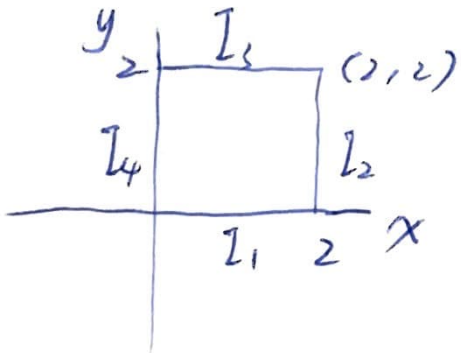
$f_{xy} = f_{yx} = 0$, $f_{xx}=2$, $f_{yy}=-2$.

$$D^2 f = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

So $(1,1)$ is a $\boxed{\text{saddle}}$ point.

$$f(x,y) = (x-1)^2-(y-1)^2$$

$$f(1,1) = 0.$$



I_1 : $x \in [0,2]$, $y=0$.

$f(x,0) = (x-1)^2 - 1$

max : 0 ($x=0$ or 2)

min : -1 ($x=1$)

I_2 : $x=2$, $y \in [0,2]$

$f(2,y) = 1-(y-1)^2$

max : 1 ($y=1$)

min : 0 ($y=0$ or 2)

I_3 : $y=2$, $x \in [0,2]$.

$f(x,2) = (x-1)^2 - 1$

max : 0 ($x=0$ or 2)

min : -1 ($x=1$)

I_4 : $x=0$, $y \in [0,2]$.

$f(0,y) = 1-(y-1)^2$

max : 1 ($y=1$)

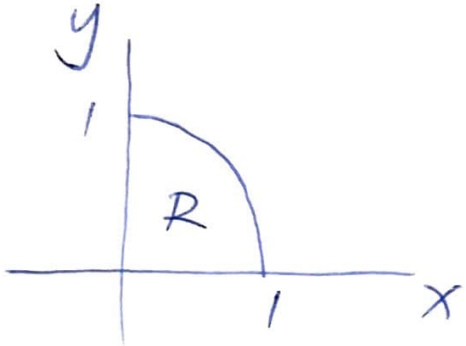
min : 0 ($y=0$ or 2)

So in R : max : 1

min : -1

$1 + (-1) = \boxed{0}$.

Q3. Solution :



In polar coordinates,

$$\boxed{R = \left\{ (r, \theta) \;\middle|\; (0 \le) r \le 1 ,\ 0 \le \theta \le \tfrac{\pi}{2} \right\}}$$

In Cartesian coordinates.

$$R = \left\{ (x, y) \;\middle|\; 0 \le x \le 1 ,\ 0 \le y \le \sqrt{1-x^2} \right\}$$

$$\left(\overset{\text{or}}{=} \left\{ (x, y) \;\middle|\; 0 \le y \le 1 ,\ 0 \le x \le \sqrt{1-y^2} \right\} \right)$$

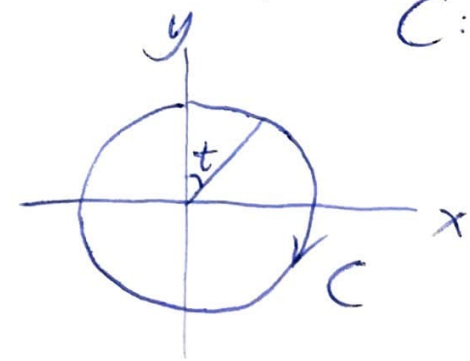
$$\iint_R 1 \, dr \, d\theta$$

$$= \iint_R \frac{1}{r} \cdot r \, dr \, d\theta$$

$$= \boxed{\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{x^2+y^2}} \, dy \, dx}$$

$$\left(\overset{\text{or}}{=} \int_0^1 \int_0^{\sqrt{1-y^2}} \frac{1}{\sqrt{x^2+y^2}} \, dx \, dy \right)$$

Q4. Solution :



$C: \vec{r}(t) = (\sin t, \cos t)$

$t \in [0, 2\pi)$

$x(t) = \sin t, \quad y(t) = \cos t.$

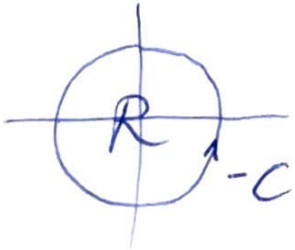
$$\int_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} (0, -\sin t)\cdot(\cos t, -\sin t)\, dt$$

$$= \int_0^{2\pi} \sin^2 t \, dt$$

$$= \int_0^{2\pi} \tfrac{1}{2}(1-\cos 2t)\, dt$$

$$= 2\pi \times \tfrac{1}{2} = \boxed{\pi}.$$

$$\int_C \vec{F}\cdot d\vec{r} = \int_C -x\, dy$$



$$= -\int_{-C} -x\, dy$$

$$= -\iint_R \left(\frac{\partial(-x)}{\partial x}\right) dx\,dy$$

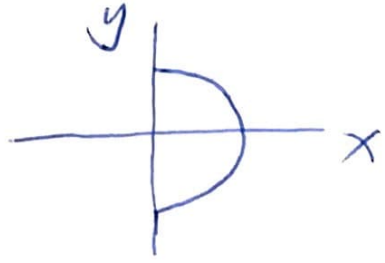
$$= \boxed{\iint_R dx\,dy}$$

$$\boxed{R = \{\underset{\text{polar}}{(r,\theta)} \mid r \le 1,\ \theta \in [0, 2\pi)\}}$$

$$\left(\overset{\text{or}}{=} \{\underset{\text{Cartesian}}{(x,y)} \mid -1 \le x \le 1,\ -\sqrt{1-x^2} \le y \le \sqrt{1-x^2}\} \right.$$

$$\left. \overset{\text{or}}{=} \{(x,y) \mid -1 \le y \le 1,\ -\sqrt{1-y^2} \le x \le \sqrt{1-y^2}\}. \right)$$

Q5. Solution:



$C: \vec{r}(t) = (\sin t, \cos t)$

$t \in [0, \pi].$

$x(t) = \sin t$, $y(t) = \cos t$. Clearly $\|\vec{r}'(t)\| = 1$. (9)

$$\text{Mass} = \int_0^{\pi} (x(t) + y(t)^2)\, dt$$

$$= \int_0^{\pi} (\sin t + \cos^2 t)\, dt$$

$$= (-\cos t)\Big|_0^{\pi} + \int_0^{\pi} \tfrac{1}{2}(1+\cos 2t)\, dt$$

$$= (-(-1) + 1) + \tfrac{1}{2} \times \pi$$

$$= \boxed{2 + \pi/2}.$$

$$10\rho \cdot \|\vec{r}'(t)\|\, \Delta t \cdot \big((0,-1) \overset{\text{dot product}}{\cdot} \vec{p}(t)\big)$$

$$= 10(x+y^2) \times 1 \times \Delta t \times \big((0,-1) \underset{\text{dot product}}{\cdot} (y-x,\ x-y)\big)$$

(10)

$$= 10(x+y^2)(y-x)\, \Delta t$$

$$\text{Work} = \int_0^{\pi} 10(\sin t + \cos^2 t) \cdot (\cos t - \sin t)\, dt$$

$$= 10 \int_0^{\pi} \sin t \cos t - \sin^2 t + \cos^3 t - \cos^2 t \sin t\, dt$$

$$= 10 \cancel{\int_0^{\pi} \sin t\, d\sin t} - 10 \int_0^{\pi} \tfrac{1}{2}(1-\cos 2t)\, dt + 10 \cancel{\int_0^{\pi} \cos^2 t\, d\sin t} + 10 \int_0^{\pi} \cos^2 t\, d\cos t.$$

$$= -10 \times \pi \times \tfrac{1}{2} + 10 \int_1^{-1} u^2\, du$$

$$= \boxed{-5\pi - 20/3}.$$